

Nneka Hamilton

CS 499

Databases

In the artifact, the story that lies behind the three databases is contained.

One of the applications that was selected for the Databases upgrade was the Traveler Gateway app. Using MongoDB and Mongoose, this application maintains data pertaining to travel destinations. The initial version of the program was only capable of retrieving travel information from the database; it did not do very many other functions. As part of this upgrade, I made modifications to the controller code that retrieves and formats trip information. These modifications made it simpler for the application to establish a connection to MongoDB. I also added other trip attributes to the program, such as photographs of the place, difficulty levels, customer ratings, and descriptive tags that add new ways to search, sort, and filter. I also added these properties to the application.

For the purpose of making the program more reliable in the event that the database is unavailable or does not contain any records, I inserted backup sample data. With this upgrade, users will be able to connect with the application, which will improve the overall process of developing and testing the application.

An further improvement that I made to the database model was the addition of more robust validation and improved data formatting. In addition, I made certain that it would be compatible with the program that is now being used. Prior to being displayed in an application, these

modifications demonstrate that database data can be transformed into content that is not only helpful but also simple to utilize.

Reflection is the seventh point.

I was able to get more knowledge about MongoDB, Mongoose, and the processing of data on the server in a full-stack web application as a result of this transition. As I became more proficient at gathering data, ensuring that it is accurate, and displaying it, I gained a deeper understanding of how to develop database-driven applications that are dependable.

The data model of the program was expanded, and fallback data was included, in order to demonstrate how essential it is to develop software that continues to function normally even in the event that it is unable to establish a connection to the database. With all of these skills under my belt, I am certain that I will be able to employ the recent database technologies to create data-driven applications that are scalable.